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Lecture 22-23 Neural Network III: Deep Learning

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. History of Deep Learning
2. Deep Neural Network
3. Main models
4. Deep feedforward networks
5. Deep Learning in the Future
6. Summary and Remark

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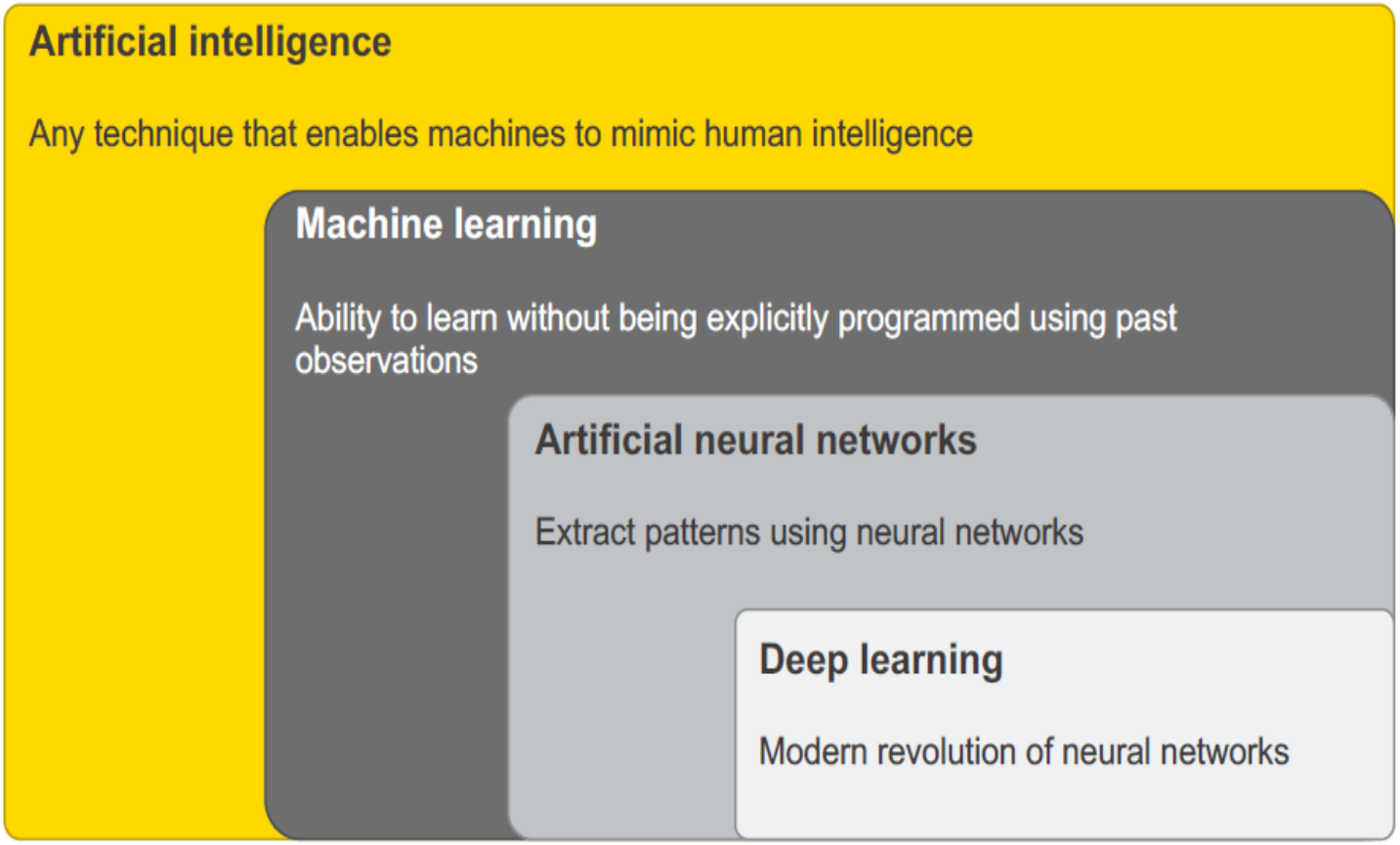
What is Deep Learning?



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History of Deep Learning

- 3 main waves in the history of neural networks/deep learning
- Wave I: 1940s-1960s cybernetics
  Some of the earliest learning algorithms we recognize today were intended to be computational models of biological learning
- Training one neuron
- Wave II: 1980-1995
  the second wave started with the connectionist approach of the period,with back-propagation (Rumelhart et al., 1986a)

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History of Deep Learning

- The central idea in connectionism is that a large number of simple computational units can achieve intelligent behavior when networked together
- During the second wave it was possible to train NN with 1-2 hidden layers
- Wave III: The current and third wave, deep learning, started around 2006 (Hinton et al., 2006; Bengio et al., 2007; Ranzato et al., 2007a)
- AlexNet: 7 layers, 650,000 neurons, 60,000,000 parameters, 630,000,000 connections, final feature layer: 4096-dimensional

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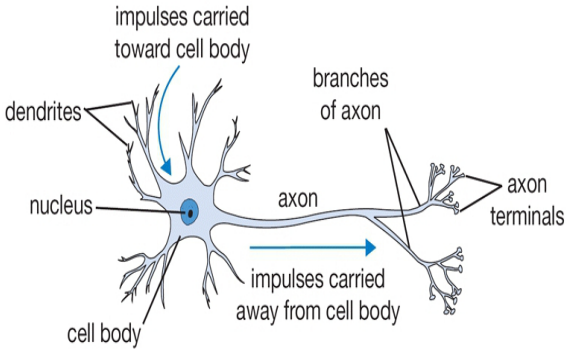
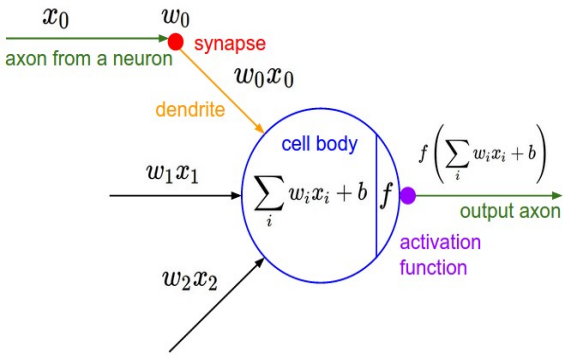
History

- Deep Learning is a Statistical Machine Learning subfield that studies Deep Neural Networks (DNN).
- DNN is a form of Neural Network (NN). NN was originated in 1943.
- NN model was inspired by biological neurons.
- Due to the lack of computational power and proper estimation techniques NN models were not well studied.



(a) Neuron



(b) Model

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Deep Learning: Formulation

- Formally, DNNs are a family of functions that can approximate any function under some conditions
- One layer DNN is simply: $Y = f(WX + b)$, where $WX + b$ is a linear regression and f is an activation function.
- Two layer DNN is defined as: $Y = f(W_2 f(W_1 X + b_1) + b_2)$
- Activation function f is a non-linear function.
- Until recently a popular choice for f was a logistic function $\sigma(z) = \frac{1}{1+e^{-z}}$ In Deep Learning literature it is referred to as *sigmoid*.
- Note that one layer DNN with $f(z) = \sigma(z)$ is the good old friend logistic regression.

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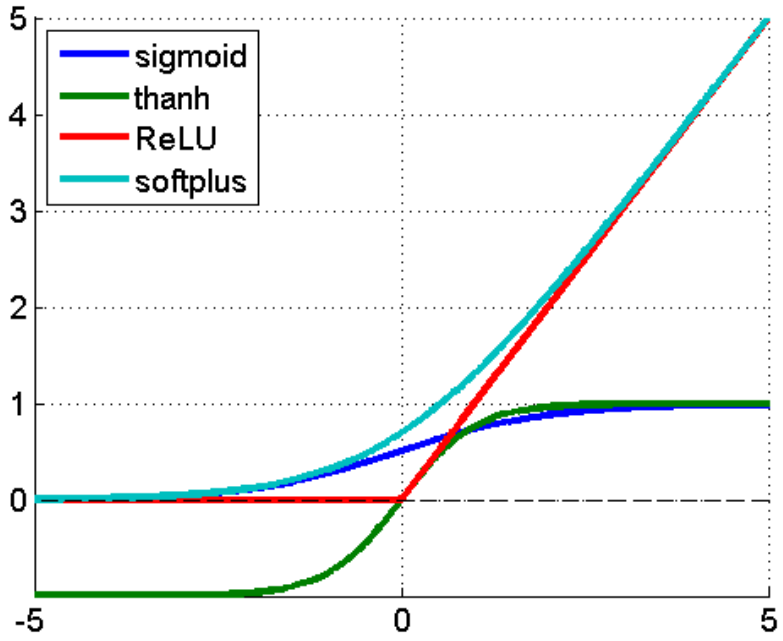
Deep Learning: Activation functions

- Currently the most popular choice for the activation function is the so called Rectified Linear Unit (ReLU): $f(z) = max(0, z)$
- Some other common choices are $tanh(z)$, $softplus(z) = \ln(1 + e^z)$, etc.



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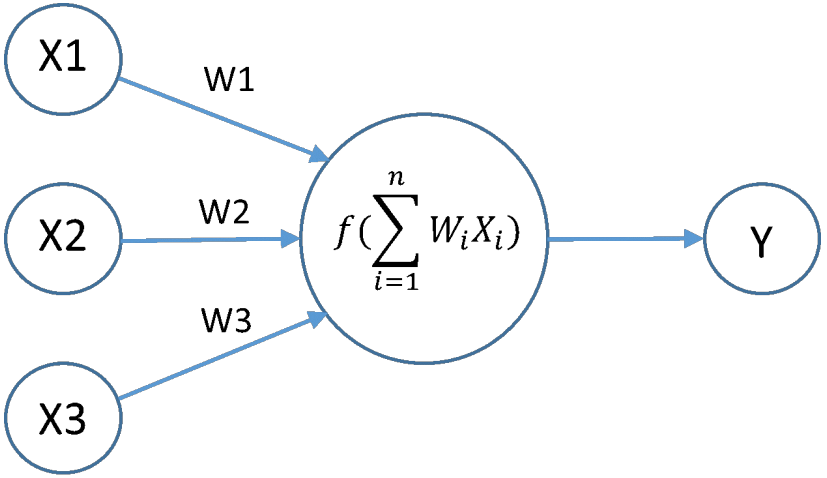
Deep Learning: Graphical representation

- Modern DNNs can have around 50 layers and the corresponding mathematical expression becomes cumbersome and useless.
- Therefore, the widely accepted way of representing DDNs is graphical



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Deep Learning: Estimation problem

- DNN, such as:

$$Y = f(W_2 f(W_1 X + b_1) + b_2)$$

- DNN is parametrized by the set of weights $W_i's$ and biases $b_i's$.
- Given a data set $\{X_i, Y_i\}_{i=1}^n$ and some loss function $L(Y_i, \hat{Y}_i)$ we'd like to estimate W_1, W_2, b_1, b_2 by minimizing $\sum_{i=1}^n L(Y_i, \hat{Y}_i)$.
- It turns out that for 2-layers networks and deeper no closed form solution is known.
- What's even worse is that the optimization problem is not convex.
- In practice parameters are estimated by different flavors of gradient descent methods.

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Artificial Neural Networks

- ANN is composed of several perceptron-like units arranged in multiple layers.
- Consists of an input layer, one or more hidden layer, and an output layer.
- Nodes in the hidden layers compute a nonlinear transform of the inputs.
- Also called a Feedforward Neural Network (since there is no backward connections between layers, viz., no loops).
- Note: All nodes between layers are assumed connected with each other.
- Advantage: Highly flexible $\rightarrow$ good performance
- Disadvantage: Black-box models not easy to interpret

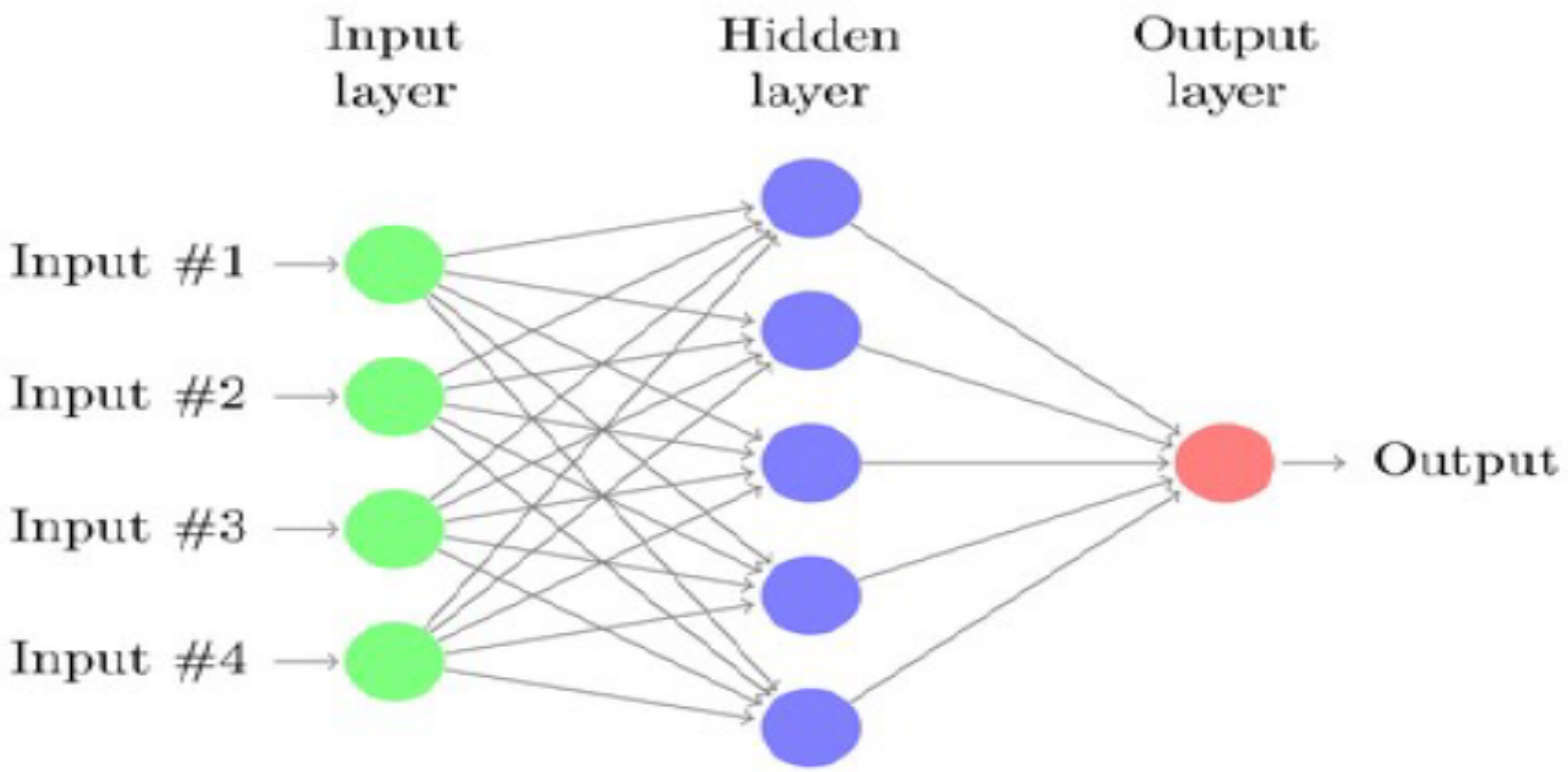

Fig: FFNN Model with one hidden layer with 5 hidden units

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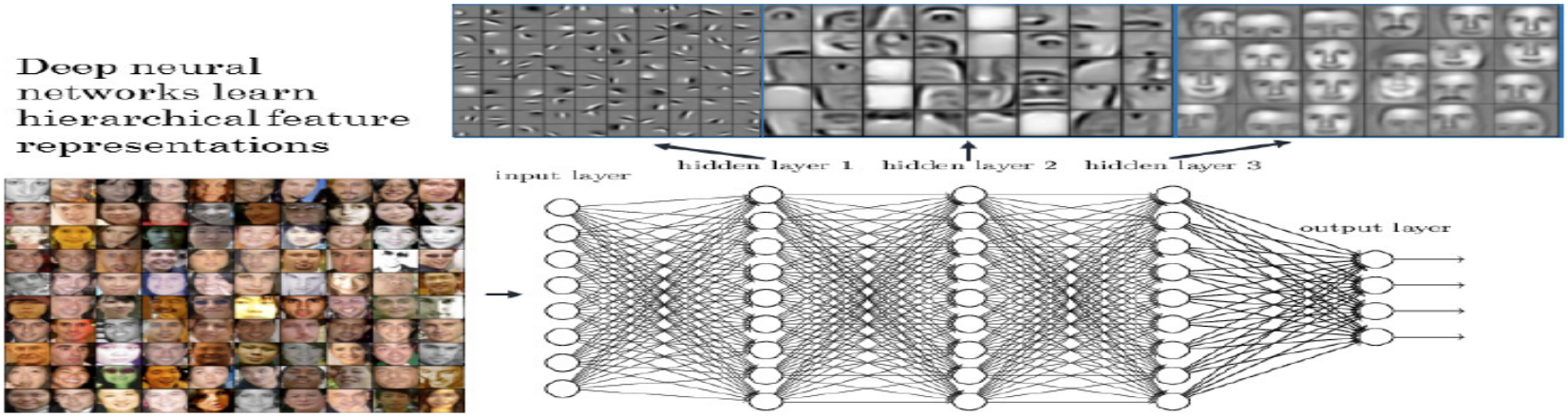
Deep feedforward networks



Slide Credit: Piyush Rai, IIT Kanpur

- also often called feedforward neural networks, or multi-layer perceptrons (MLPs), are the quintessential deep learning models
- Hidden layers can automatically extract features from data.
- The bottom-most hidden layer captures very low level features (e.g., edges). Bottom Most = First Layer
- Subsequent hidden layers learn progressively more high-level features (e.g., parts of objects) that are composed of previous layer's features.

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Example Problem

Will I pass this class?

- Let's start with a simple two feature model
- x_1 = Number of lectures you attend
- x_2 = Hours spent on the final project

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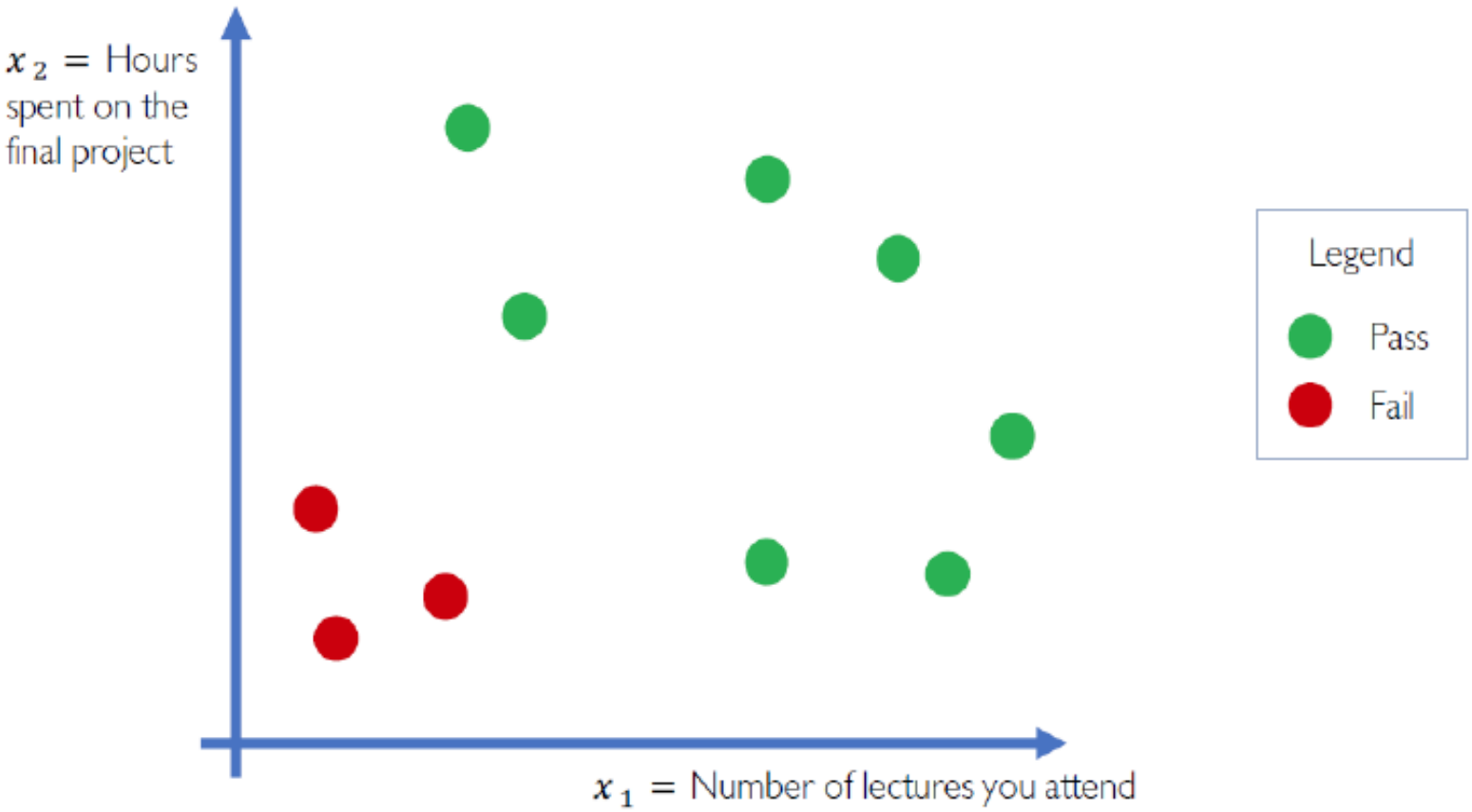
Example Problem: Will I pass this class?



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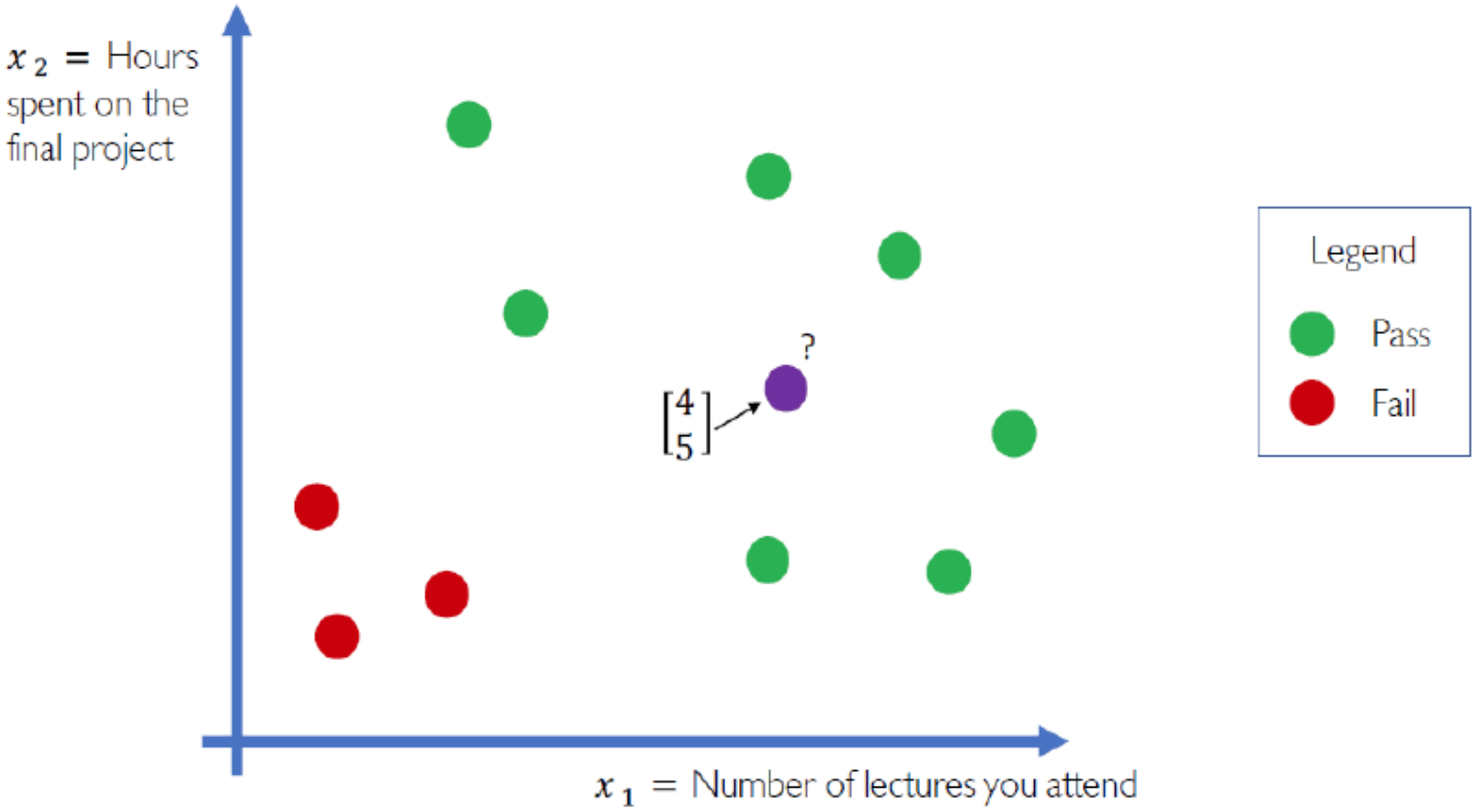

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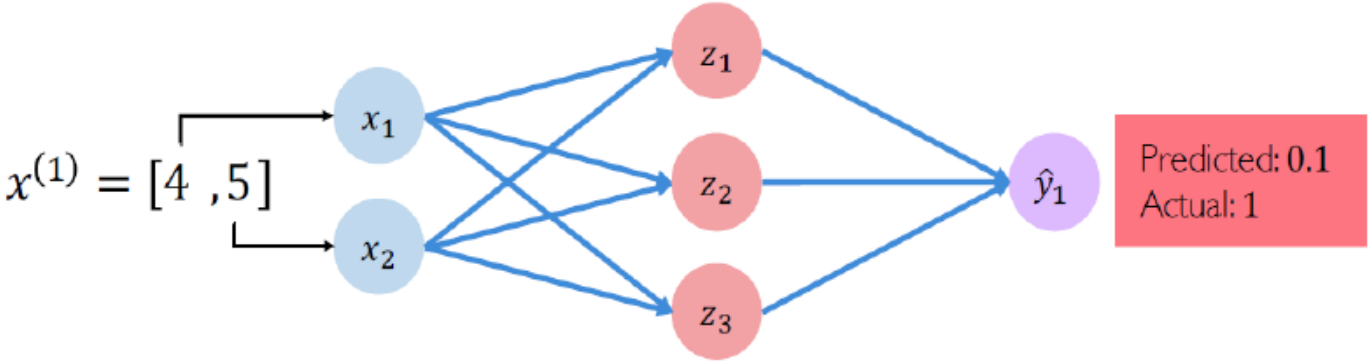
Example Problem: Will I pass this class?



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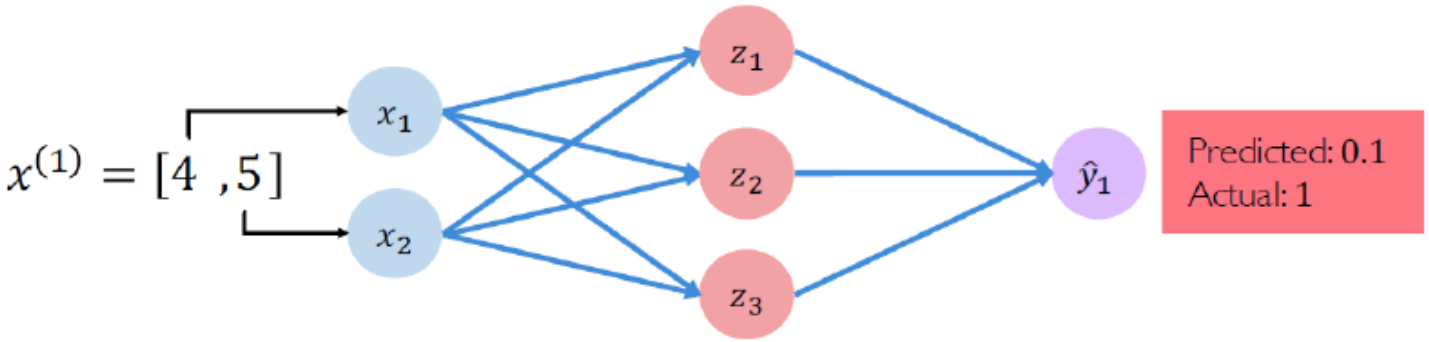
Quantifying Loss

*The **loss** of our network measures the cost incurred from incorrect predictions*



$$\mathcal{L}\big(\underbrace{f\left(x^{(i)};\boldsymbol{\theta}\right)}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}}\big)$$

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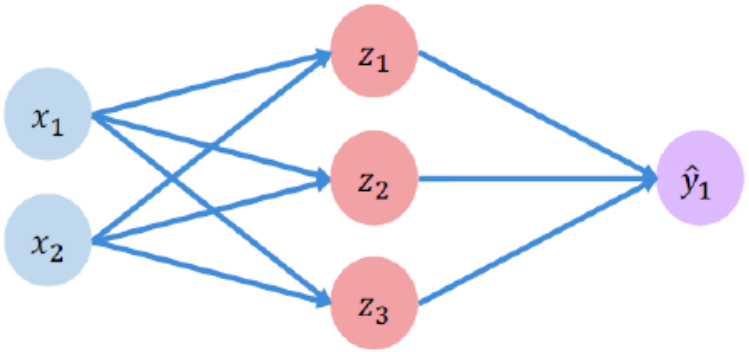
Empirical Loss

*The **empirical loss** measures the total loss over our entire dataset*

$$x = \begin{bmatrix} 4, & 5 \\ 2, & 1 \\ 5, & 8 \\ \vdots & \vdots \end{bmatrix}$$



$$\begin{array}{cc} f(x) & y \\ \begin{bmatrix} 0.1 \\ 0.8 \\ 0.6 \\ \vdots \end{bmatrix} & \begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \end{bmatrix} \end{array}$$

Also known as:
- Objective function
- Cost function
- Empirical Risk

$$J(\theta) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}\big(\underbrace{f(x^{(i)};\theta)}_{\text{Predicted}}, \underbrace{y^{(i)}}_{\text{Actual}}\big)$$

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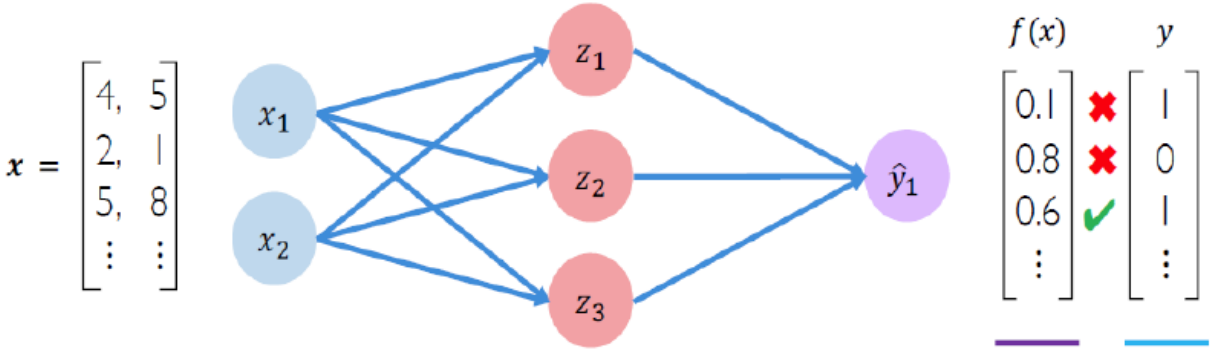
Binary Cross Entropy Loss

Cross entropy loss *can be used with models that output a probability between 0 and 1*



$$J(\theta) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{y^{(i)}}_{\text{Actual}} \log\Big(\underbrace{f\big(x^{(i)};\theta\big)}_{\text{Predicted}}\Big) + (1-\underbrace{y^{(i)}}_{\text{Actual}}) \log\Big(1-\underbrace{f\big(x^{(i)};\theta\big)}_{\text{Predicted}}\Big)$$

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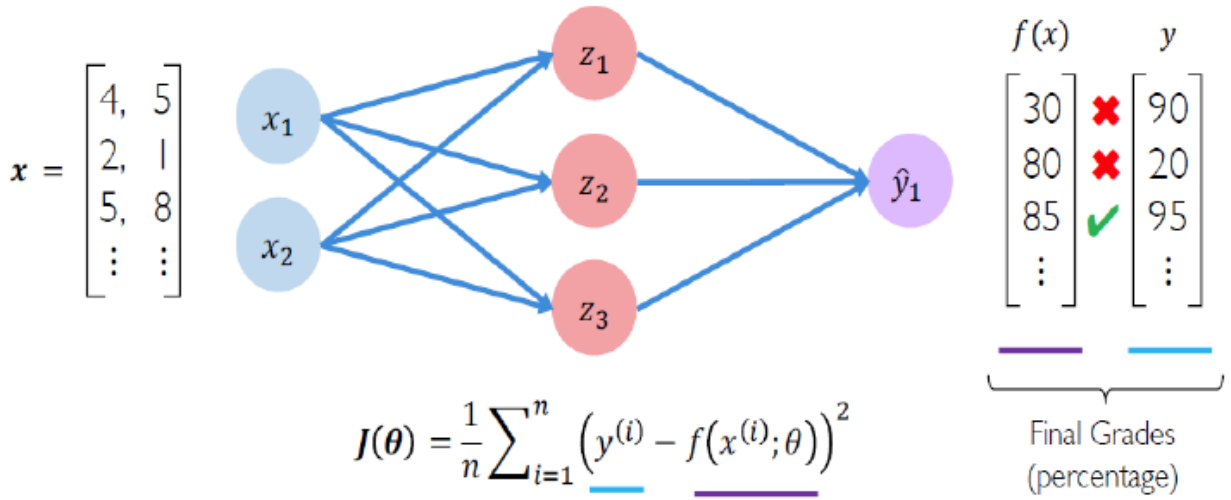
Mean Squared Error Loss

Mean squared error loss *can be used with regression models that output continuous real numbers*



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Deep feedforward networks - basic

- Feedforward networks are the first type of deep networks to study
- Feedforward means that the signal is not refer to the previous layer
- Two consecutive layers are fully connected. Unimportant connections should get less weight during training
- Deep means more than 1 hidden layer
- Otherwise the topology of the network is a hyperparamter. The data scientist has to choose how wide the layers should be
- Recent experimental studies revealed that increasing depth is more important than width of the network

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Deep feedforward networks - theory

- Deep neural networks are regarded as universal function approximators
- Deeper networks yield better approximations
- However deep networks have more parameters to train (estimate), which requires more data
- Due to complexity of the neural networks there is no exact answer to how much data do we need to train a given network
- Impossible (at least now) to construct prediction intervals

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Deep feedforward networks - Practice

- With not much theory in hand on how to exactly design the network one has to experiment
- Since there is no theoretical measure of accuracy for deep networks we again have to resort to experiments
- In practice in order to compare models researchers cross-validate them using one fixed data set
- Popular data sets: MNIST, ImageNet etc
- The fact that deep nets are not well understood does not imply they are scientifically unsound

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Deep feedforward networks - software

- The deep nets are complicated
- However implementing a network is not that hard since there exists a lot of free libraries
- Obtaining good results is hard
- At a basic level you only need to specify the graph: nodes and connections and choose the training method
- `Pytorch` is one of the machine learning libraries available in Python capable of deep learning
- It is capable of running quite large networks

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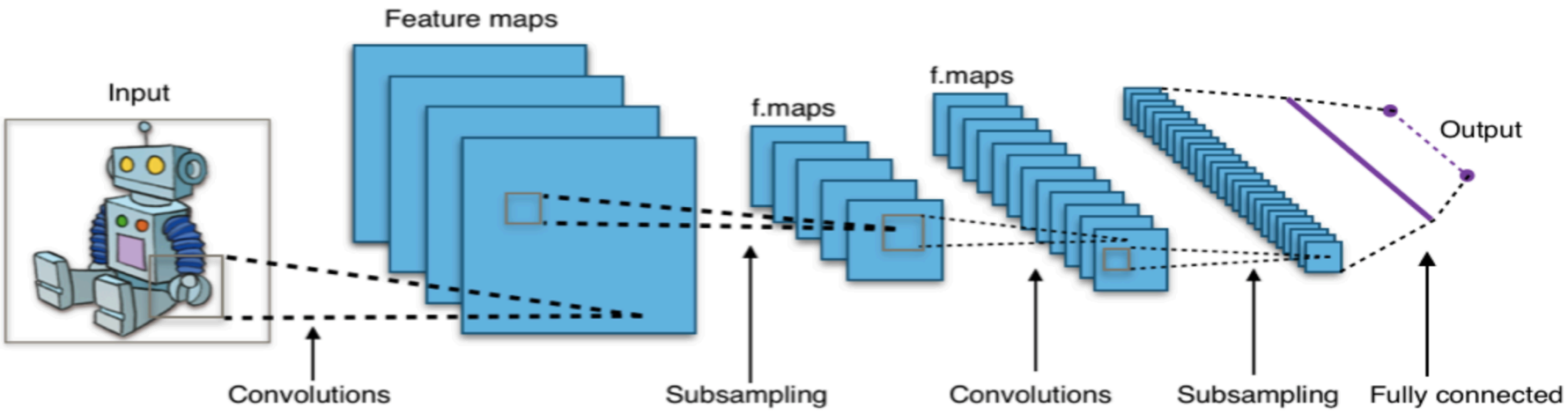
Convolutional Networks



- Convolutional neural networks (CNN) evolved mainly to deal with images.
- Convolutional neural networks (CNN) are simply neural networks that use convolution in place of general matrix multiplication in at least one of their layers.
- One of the key features is the use of shared weight in convolutional layers.

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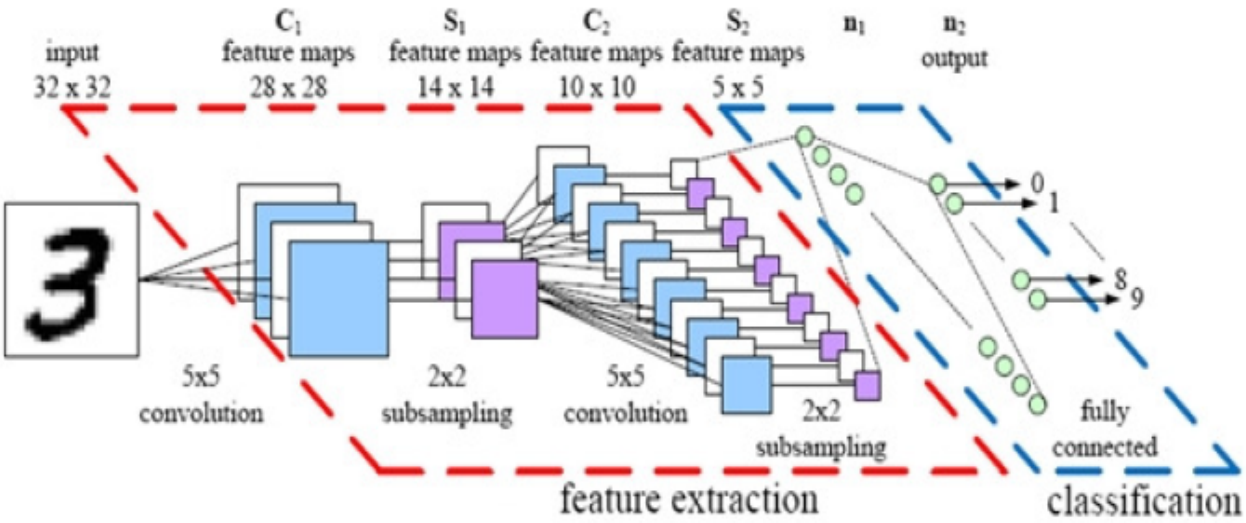
- Convolution: Extract "local" properties of the signal. Uses a set of "filters" that have to be learned (these are the "weight" W between layers).
- Pooling: Downsamples the outputs to reduce the size of representation.
- Note: A nonlinearity is also introduced after the convolution layer.



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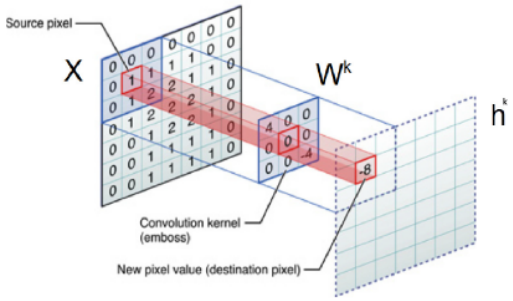
Convolution

- An operation that captures local (e.g., spatial) properties of a signal:



- Mathematically, the operation is defined as

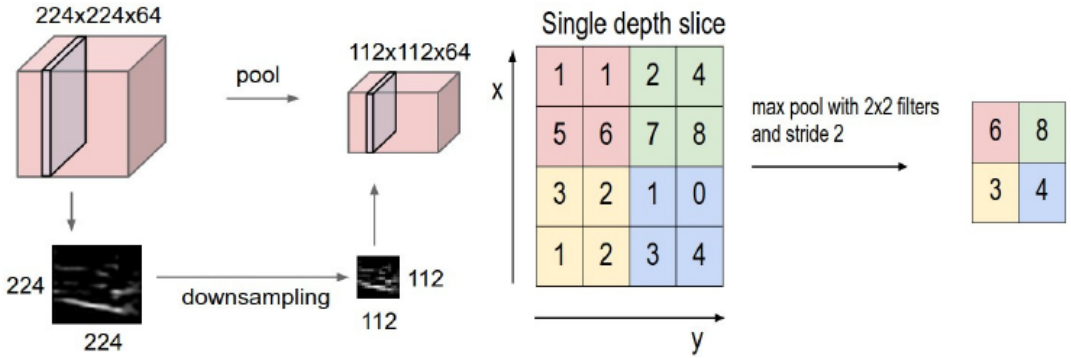
$$h_{ij}^k = f((W^k * X)_{ij} + b_k),$$

where W^k is a filter, $*$ is the convolution operator, and f is a nonlinearity.

- Usually a number of filters $\{W^k\}_{k=1}^K$ are applied (each will produce a separate "feature map"). These filters need to be learned.
- Size of these filters also need to be specified.

Pooling / Subsampling

- This operation is used to reduce the size of the representation.



- Max-pooling is commonly used (replacing a block of values by the max value).

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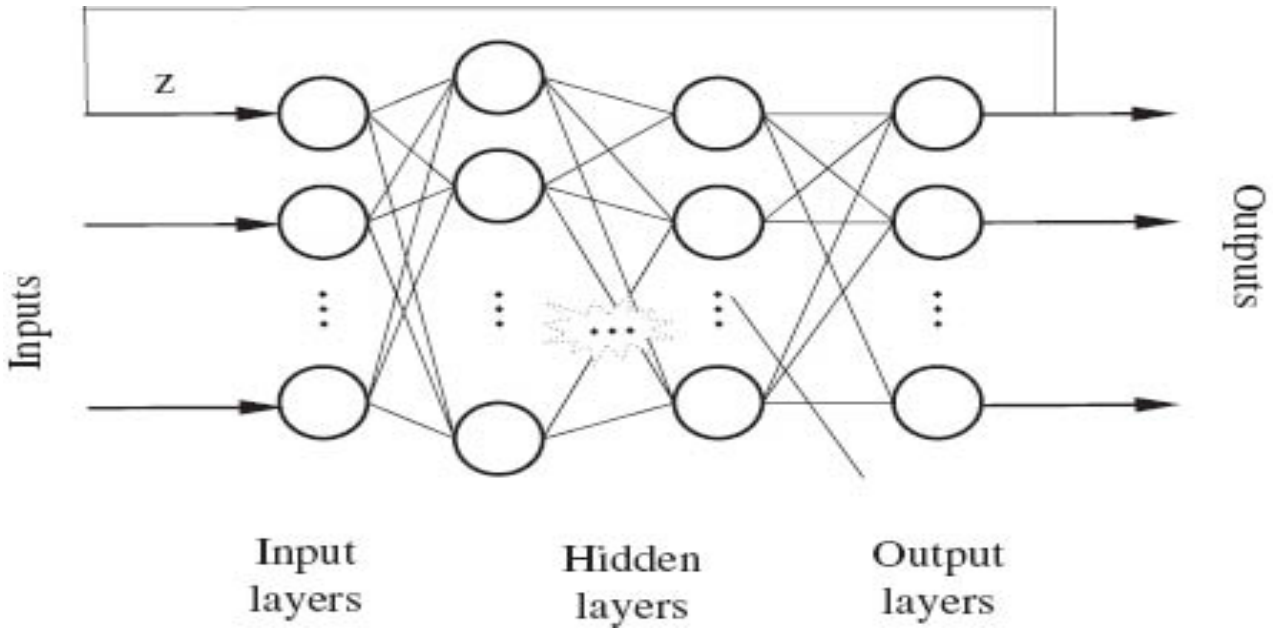
Recurrent Neural Networks



- Recurrent neural networks or RNNs are a family of neural networks for handling sequential data
- One of the key features is that connections between units form a directed cycle, which allows it to exhibit dynamic temporal behavior.

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Recurrent Neural Networks

- Recurrent neural networks (RNNs) are networks with loops, allowing information to persist

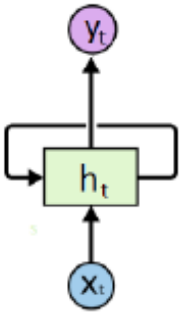
$$\boxed{h_t} = \boxed{f_W}(\boxed{h_{t-1}}, \boxed{x_t})$$

new state

old state

input vector at some time step

some function with parameters W



- Have memory that keeps track of information observed so far
- Maps from the entire history of previous inputs to each output
- Handle sequential data

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Deep Learning in the Future

- Facebook hired Yann LeCun; Google hired Geoffrey Hinton; and Baidu hired Andrew Ng.
- Automatic driving car - George Hotz is taking on Google and Tesla by himself (Bloomberg Businessweek)
- Deep Blue versus Kasparov (IBM) - hardware rather than software, can do in a personal laptop nowadays
- Alphago (Google deepmind) - Human Go champion 'speechless' after 2nd loss to Google DeepMind AlphaGo computer (CBC news).
- A main criticism of deep learning concerns the lack of theory surrounding many of the methods.

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Summary and Remark

- In practice training of a network can take weeks
- It is very common to use graphics cards for training (specifically CUDA on NVIDIA GPUs)
- state of the art libraries are Theano (Python), Torch (lua), Caffe (C++), Tensor Flow.